

Vector Space Models & Word Embeddings

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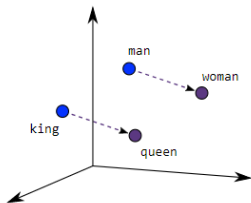


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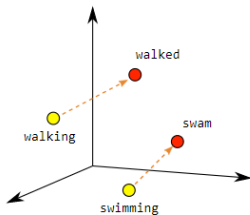


LMH
Lady Margaret Hall
University of Oxford

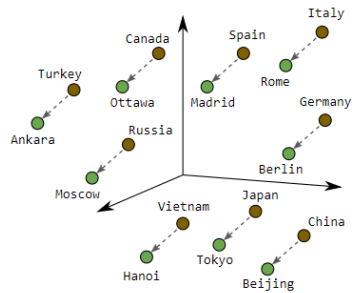
June 21, 2026



Male-Female



Verb Tense



Country-Capital

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- ▶ Text is symbolic; machines require numerical representations to process it.
- ▶ Classical NLP struggled with sparse, high-dimensional vectors.
- ▶ Vector Space Models (VSMs) and Word Embeddings represent words in a continuous vector space.
- ▶ These models capture semantic meaning and relationships geometrically.
- ▶ Basis for modern NLP methods including Transformers.

- ▶ Understand and implement basic Vector Space Models.
- ▶ Differentiate between word-by-word and word-by-document designs.
- ▶ Use similarity measures to compute semantic relatedness.
- ▶ Explain the rationale behind One-Hot vs. Dense word vectors.
- ▶ Understand and implement basic Word2Vec models (CBOW and Skip-gram).
- ▶ Appreciate the improvements of GloVe and fastText over earlier models.

Vector Space Model: **Introduction**

Why learn vector space models?

Where are you heading?

Where are you from?



Different meaning

What is your age?

How old are you?



Same Meaning

- ▶ Words and sentences can have different meanings depending on context.
- ▶ Vector space models help capture semantic similarity and differences.
- ▶ Useful for tasks like paraphrase detection, question answering, and information retrieval.

Firth, 1957

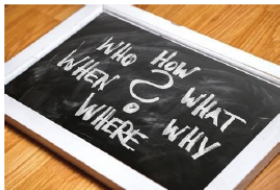


(Firth, J. R. 1957:11)

- ▶ VSM represents words or documents as vectors in an n -dimensional space.
- ▶ Based on the **distributional hypothesis**: Words that occur in similar contexts have similar meanings.
- ▶ Forms the foundation for information retrieval, document classification, and word embeddings.

- ▶ A **term-document matrix**: Rows represent words (terms), columns represent documents (or vice versa).
- ▶ Each cell contains a value such as:
 - Term Frequency (TF)
 - TF-IDF (Term Frequency-Inverse Document Frequency)
 - Co-occurrence count
- ▶ The matrix is typically high-dimensional and sparse:
 - Size = $|\text{Vocabulary}| \times |\text{Documents}|$

- ▶ You eat *cereal* from a *bowl* → Capturing semantic similarity (paraphrase understanding)
- ▶ You *buy* something and someone else *sells* it → Capturing relational meaning (analogies)



Information Extraction



Machine Translation



Chatbots

Word-by-Word vs Word-by-Document Designs

- ▶ Co-occurrence $\rightarrow$ Vector representation
- ▶ Relationships between words/documents

Number of times they *occur together within a certain distance k*

$k=2$

I like simple data

I prefer simple raw data

	simple	raw	like	I
data	2	1	1	0

n

Number of times a word *occurs within a certain category*

	Corpus		
	Entertainment	Economy	Machine Learning
data	500	6620	9320
film	7000	4000	1000

► Word-by-Word:

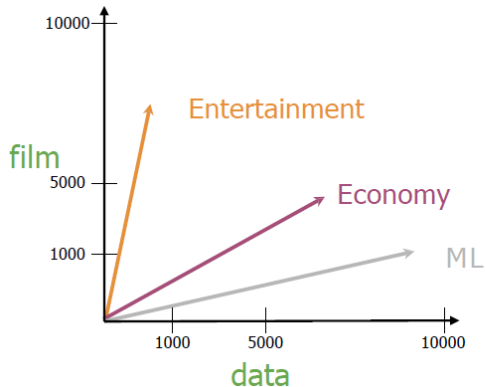
- Matrix built from co-occurrence counts between words.
- Captures semantic similarity directly.
- Better for word similarity tasks.

► Word-by-Document:

- Matrix built from word frequencies in documents.
- Useful for document classification and search.
- Better for document-level tasks.

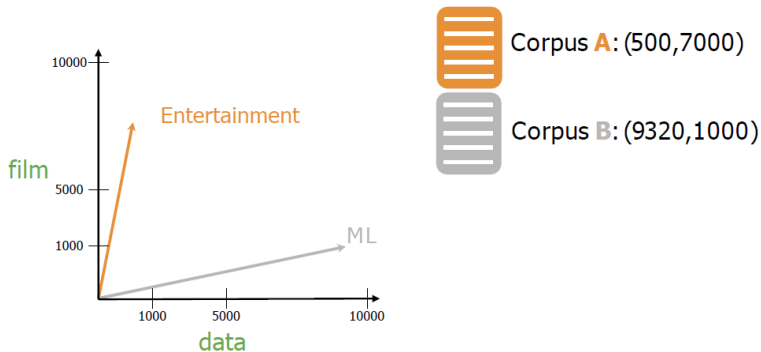
Tradeoffs: Choice depends on the task: word similarity vs. document analysis.

Similarity Measures



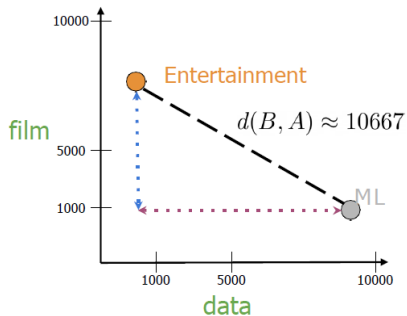
	Entertainment	Economy	ML
data	500	6620	9320
film	7000	4000	1000

Measures of "similarity:"
Angle
Distance



- ▶ Measures the straight-line distance between two points in space.
- ▶ Formula: $d(\mathbf{x}, \mathbf{y}) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$
- ▶ Sensitive to the scale (magnitude) of the vectors.

Euclidean Distance (cont.)



Corpus **A**: (500,7000)



Corpus **B**: (9320,1000)

$$d(B, A) = \sqrt{(B_1 - A_1)^2 + (B_2 - A_2)^2}$$
$$c^2 = a^2 + b^2$$

$$d(B, A) = \sqrt{(-8820)^2 + (6000)^2}$$

Euclidean Distance for n-dimensional vectors

	data	$\vec{w}$ boba	$\vec{v}$ ice-cream
AI	6	0	1
drinks	0	4	6
food	0	6	8

$$= \sqrt{(1 - 0)^2 + (6 - 4)^2 + (8 - 6)^2}$$

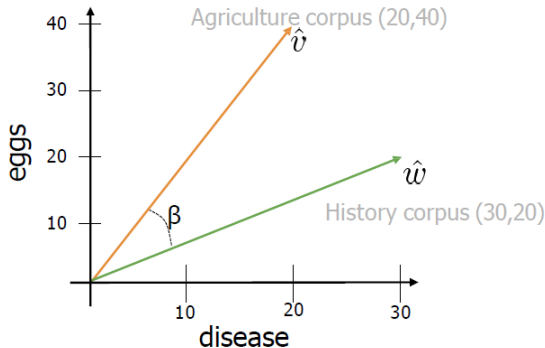
$$= \sqrt{1 + 4 + 4} = \sqrt{9} = 3$$

$$d(\vec{v}, \vec{w}) = \sqrt{\sum_{i=1}^n (v_i - w_i)^2} \longrightarrow \text{Norm of } (\vec{v} - \vec{w})$$

```
# Create numpy vectors v and w
v = np.array([1, 6, 8])
w = np.array([0, 4, 6])

# Calculate the Euclidean distance d
d = np.linalg.norm(v-w)
# Print the result
print("The Euclidean distance between v and w is: ", d)
```

The Euclidean distance between v and w is: 3



$$\hat{v} \cdot \hat{w} = \|\hat{v}\| \|\hat{w}\| \cos(\beta)$$

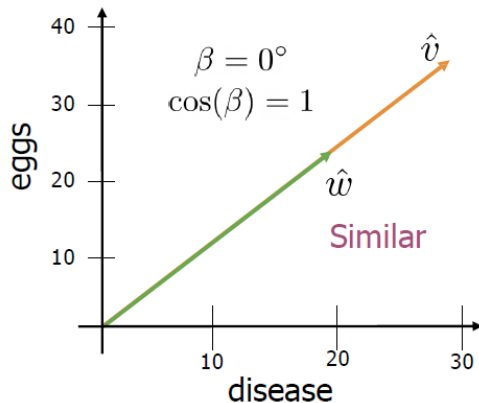
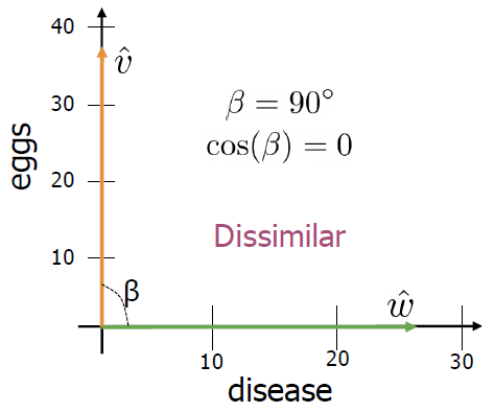
$$\cos(\beta) = \frac{\hat{v} \cdot \hat{w}}{\|\hat{v}\| \|\hat{w}\|}$$

$$= \frac{(20 \times 30) + (40 \times 20)}{\sqrt{20^2 + 40^2} \times \sqrt{30^2 + 20^2}} = 0.87$$

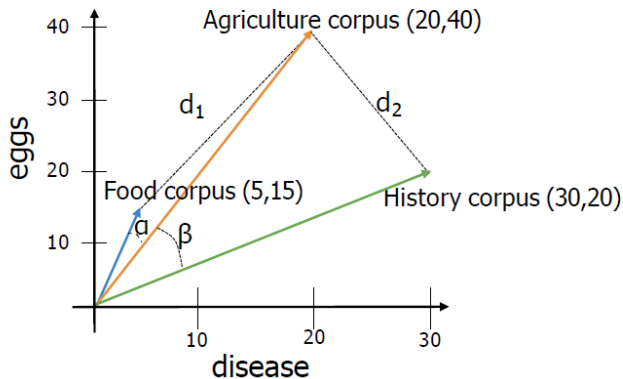
- Measures the cosine of the angle between two vectors.
- Formula: $\text{cosine}(\mathbf{x}, \mathbf{y}) = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|}$
- Ranges from -1 (opposite) to 1 (same direction).

- ▶ Less sensitive to magnitude, focuses on orientation.

Cosine Similarity (cont.)



- Cosine similarity when corpora are different sizes



Euclidean distance: $d_2 < d_1$

Angles comparison: $\beta > \alpha$

The cosine of the angle
between the vectors

Manipulating word vectors



USA



Washington
DC

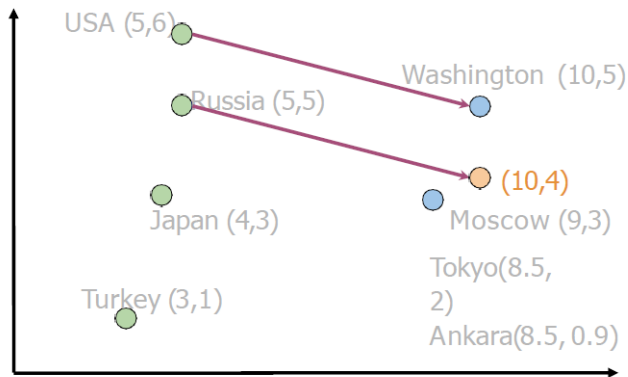


Russia



?

Manipulating word vectors (cont.)



$$\text{Washington} - \text{USA} = \begin{bmatrix} 5 & -1 \end{bmatrix}$$

$$\text{Russia} + \begin{bmatrix} 5 & -1 \end{bmatrix} = \begin{bmatrix} 10 & 4 \end{bmatrix}$$



Moscow

[Mikolov et al, 2013, Distributed Representations of Words and Phrases and their Compositionality]

	$d > 2$		
oil	0.20	...	0.10
gas	2.10	...	3.40
city	9.30	...	52.1
town	6.20	...	34.3

How can you visualize if your representation captures these relationships?



oil & gas



town & city

Visualization of word vectors (cont.)

$d > 2$

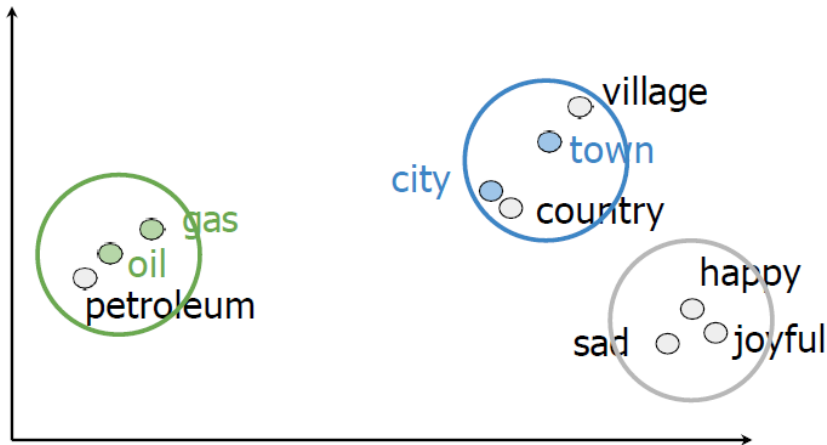
oil	0.20	...	0.10
gas	2.10	...	3.40
city	9.30	...	52.1
town	6.20	...	34.3

PCA →

$d = 2$

oil	2.30	21.2
gas	1.56	19.3
city	13.4	34.1
town	15.6	29.8

Visualization of word vectors (cont.)





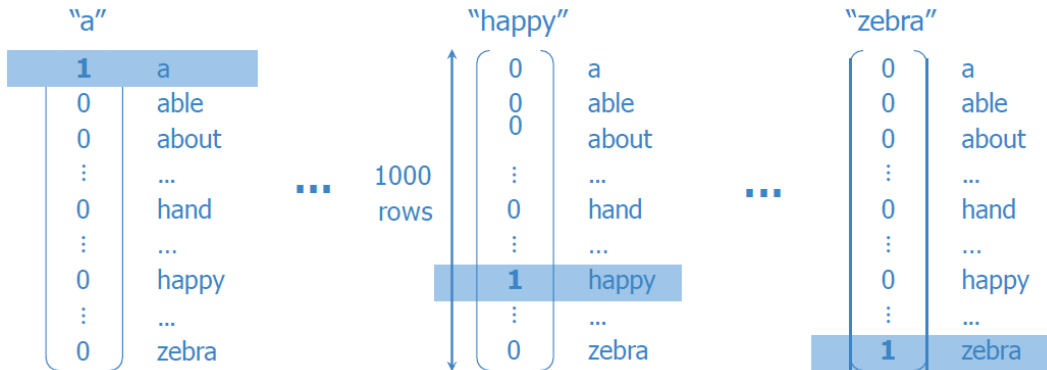
Word Embedding: **One-Hot vs. Dense Vectors**

Word	Number
a	1
able	2
about	3
...	...
hand	615
...	...
happy	621
...	...
zebra	1000

- + Simple
- Ordering: little semantic sense

hand 615 < happy 621 < zebra 1000
?! ?!

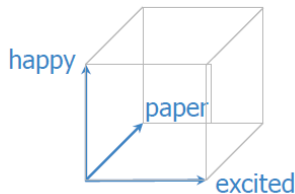
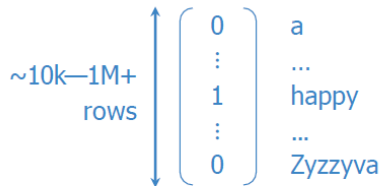
One-hot vectors



One-hot vectors (cont.)

Word	Number			
a	1	1	$\begin{pmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix}$	a
able	2	2		able
about	3	3		about
...	...	...		...
hand	615	615		hand
...	...	...		...
happy	621	621		happy
...	...	...		...
zebra	1000	1000		zebra

- + Simple
- + No implied ordering
- Huge vectors
- No embedded meaning

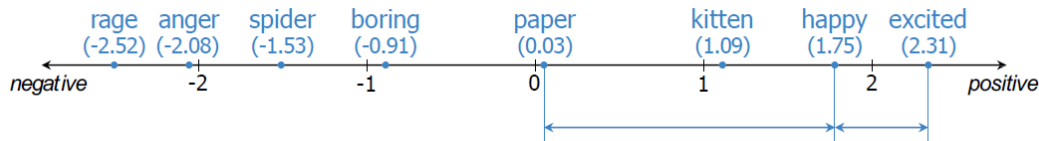


$$\begin{aligned} d(\text{paper}, \text{excited}) \\ &= d(\text{paper}, \text{happy}) \\ &= d(\text{excited}, \text{happy}) \end{aligned}$$

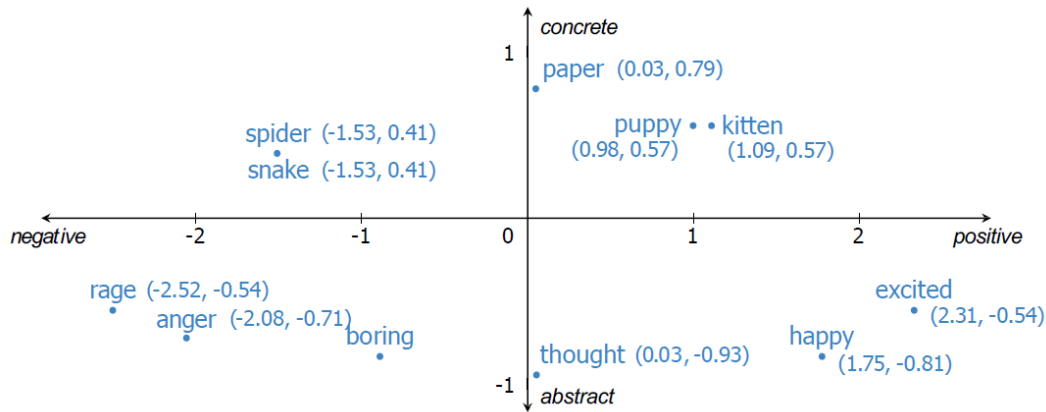
- ▶ Each word is represented as a unique vector.
- ▶ Vector has 1 at one position, 0 elsewhere.
- ▶ **Problems:**
 - High-dimensional and sparse.
 - No semantic meaning or similarity captured.

- ▶ Vectors are learned from data.
- ▶ Low-dimensional (e.g., 100–300).
- ▶ Capture semantic relationships.
- ▶ Example: "king" – "man" + "woman" $\approx$ "queen"

Meaning as vectors



Meaning as vectors (cont.)



+ Low dimension

+ Embed meaning

- e.g. semantic distance

forest $\approx$ tree forest $\not\approx$ ticket

- e.g. analogies

Paris:France :: Rome:?

"happy"

$\sim 100 \text{---} \sim 1000$
rows

$$\begin{pmatrix} 0.123 \\ \vdots \\ -4.059 \\ \vdots \\ 1.891 \end{pmatrix}$$

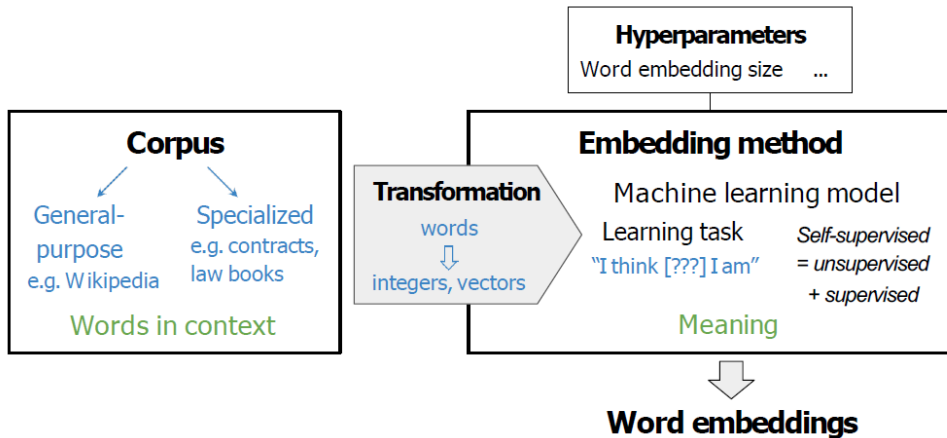
word vectors

integers

one-hot vectors

word embedding vectors

"word vectors"
word embeddings



Embedding Method: **Word2Vec**

- ▶ Dense vector representations of words learned from context.
- ▶ Encode syntactic and semantic meaning.
- ▶ Vectors reflect relationships like synonymy, analogy, etc.

- ▶ **word2vec** (Google, 2013)
 - Continuous bag-of-words (CBOW)
 - Continuous skip-gram / Skip-gram with negative sampling (SGNS)
- ▶ **Global Vectors (GloVe)** (Stanford, 2014)
- ▶ **fastText** (Facebook, 2016)
 - Supports out-of-vocabulary (OOV) words

- ▶ Deep learning-based, contextual embeddings:
 - **BERT** (Google, 2018)
 - **ELMo** (Allen Institute for AI, 2018)
 - **GPT-2** (OpenAI, 2018)
- ▶ Tunable pre-trained models available

- ▶ Word2Vec learns word embeddings from large text corpora.
- ▶ Two main architectures:
 - Continuous Bag-of-Words (CBOW)
 - Skip-gram
- ▶ Both models use neural networks to predict words based on context.

- ▶ **Goal:** Predict target word from context.
- ▶ **Architecture:**
 - **Input:** Surrounding words (context)
 - **Output:** Target word
- ▶ Fast and accurate for frequent words.
- ▶ **Example:**
 - Context: “the ___ sat on the mat” → Predict “cat”

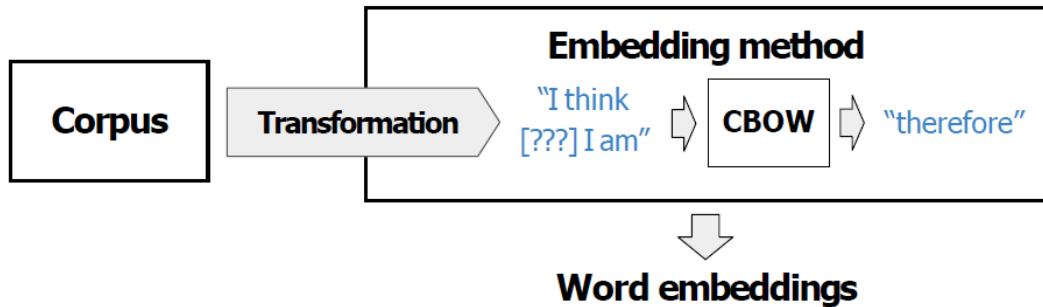
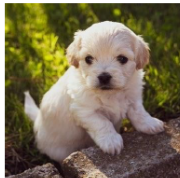


Figure 2: CBOW Architecture



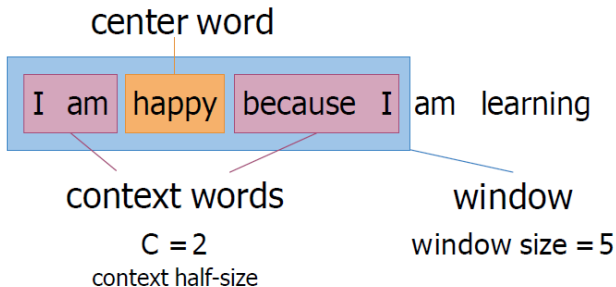
The little _____ is barking

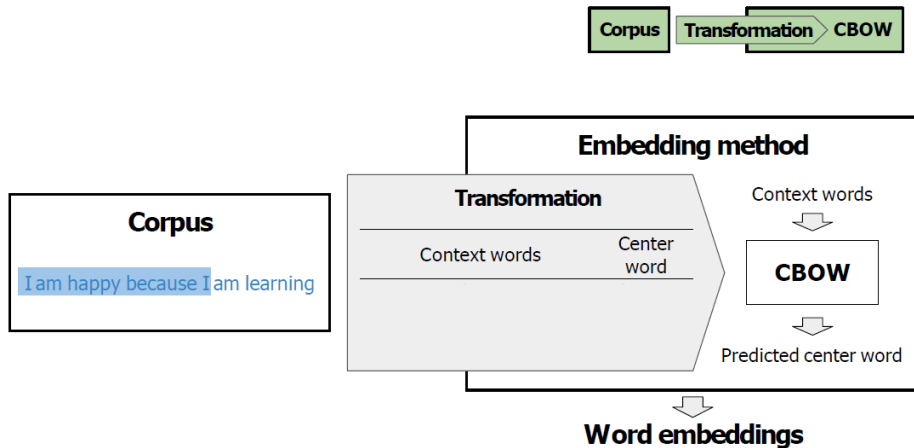


dog
puppy
hound
terrier

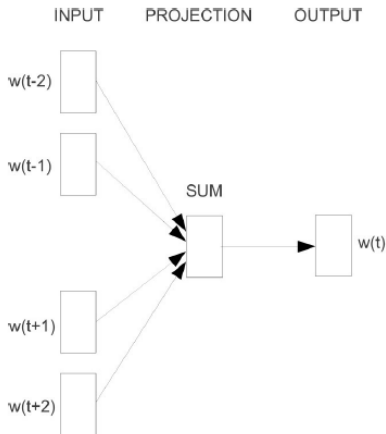
...

Creating a training example



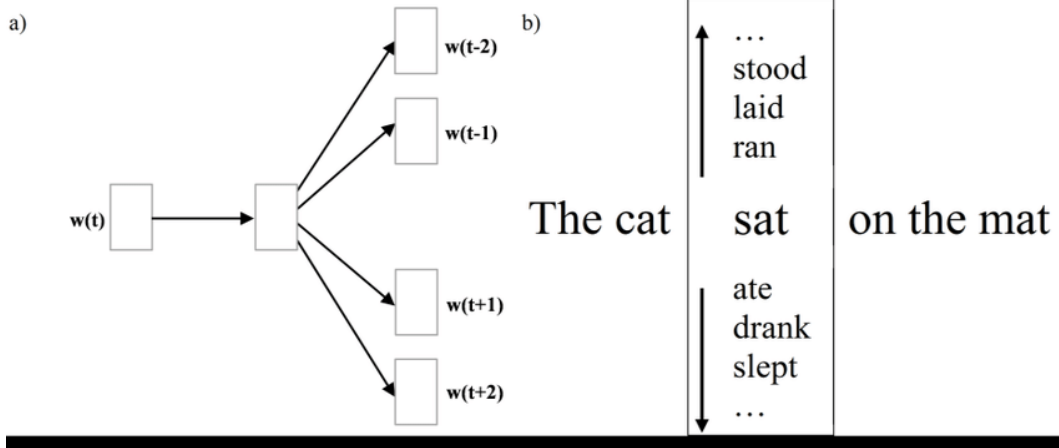


- ▶ **Goal:** Predict context words from a target word.
- ▶ **Architecture:**
 - **Input:** Target word
 - **Output:** Surrounding words (context)
- ▶ Better for rare words.
- ▶ **Example:**
 - Input: “cat” → Output: “the”, “sat”, “on”, “the”, “mat”



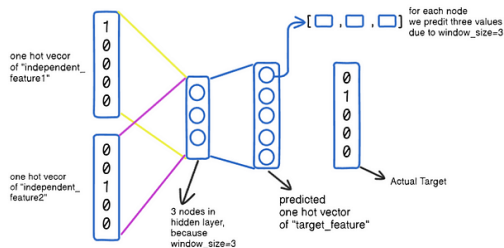
Source: Mikolov, T., Chen, K.,
Corrado, G.S., & Dean, J. (2013).
[Efficient Estimation of Word
Representations in Vector Space](#)

Figure 3: Skip-Gram Architecture

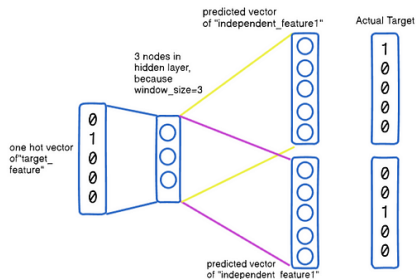


Word2vec

CBOW



Skip-gram



- ▶ Does not consider sub-word information.
- ▶ Same vector for all senses of a word (polysemy issue).
- ▶ Ignores word order within the context window.

GloVe – Global Vectors

Proposed by: Pennington et al., 2014

- ▶ Combines global matrix factorization with local context windows.
- ▶ Captures co-occurrence statistics of words across entire corpus.
- ▶ Embeddings reflect ratios of co-occurrence probabilities.

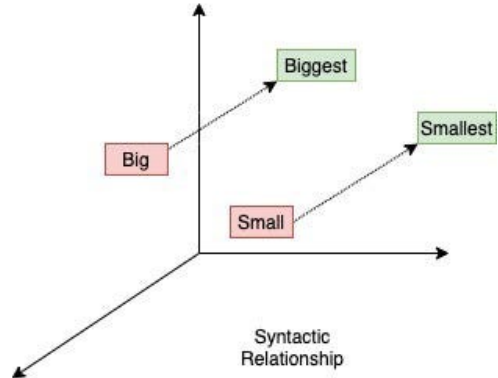
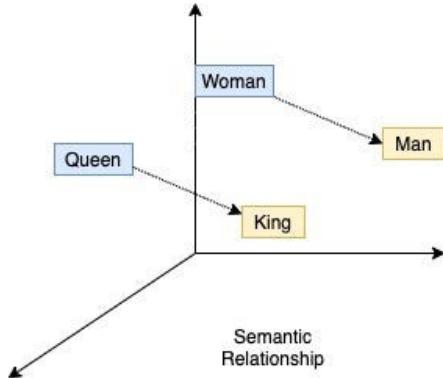
Loss function:

$$J = \sum_{i,j=1}^V f(P_{ij}) \left(w_i^\top \tilde{w}_j + b_i + \tilde{b}_j - \log X_{ij} \right)^2$$

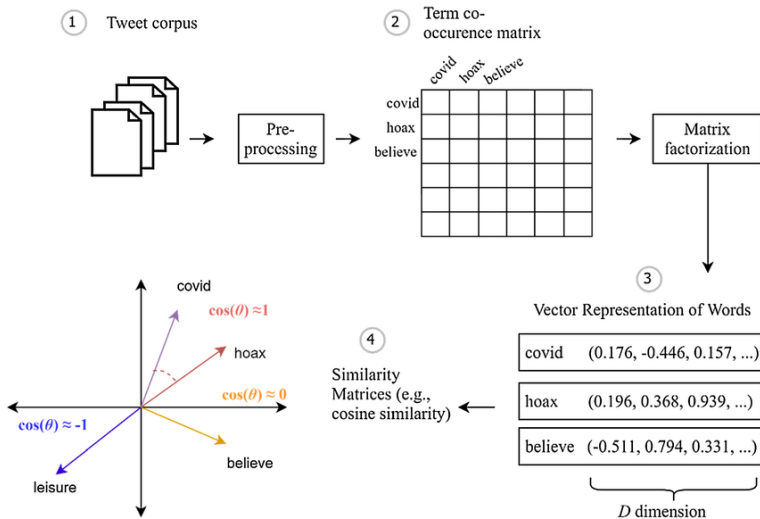
where:

- ▶ X_{ij} : number of times word j occurs in the context of word i
- ▶ P_{ij} : probability of word j in the context of i
- ▶ $w_i, \tilde{w}_j$: word and context word vectors
- ▶ $b_i, \tilde{b}_j$: bias terms
- ▶ f : weighting function

GloVe – Global Vectors (cont.)



GloVe – Global Vectors (cont.)



fastText – Subword Embeddings

- ▶ Developed by Facebook AI (2016)
- ▶ Builds on Word2Vec by incorporating character n-grams
- ▶ Useful for morphologically rich languages and rare words
- ▶ Better handling of OOV (out-of-vocabulary) words

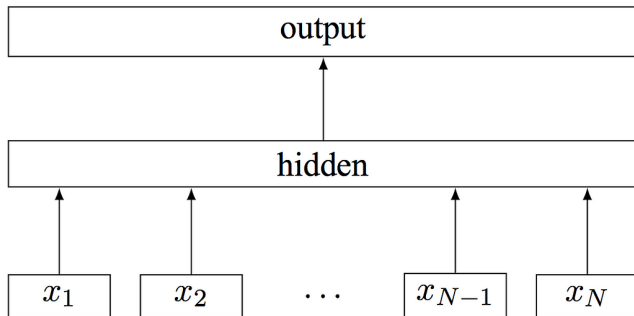
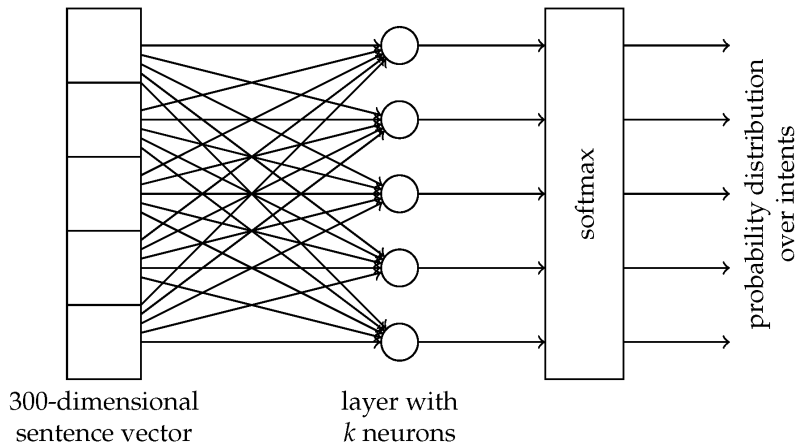


Figure 1: Model architecture of `fastText` for a sentence with N ngram features $x_1, \dots, x_N$. The features are embedded and averaged to form the hidden variable.



FastText-Based Intent Detection for Inflected Languages

- ▶ Word: playing
- ▶ Character n-grams ($n=3$): ["pla", "lay", "ayi", "yin", "ing"]
- ▶ Word vector = sum of n-gram vectors

Comparison: Word2Vec vs. GloVe vs. fastText

Feature	Word2Vec	GloVe	fastText
Local Context	✓	—	✓
Global Info	—	✓	✓ (partial)
Subword Info	—	—	✓
Handles OOV	—	—	✓

► High Dimensionality:

- Vectors can become very high-dimensional, leading to computational inefficiency.
- Curse of dimensionality: distance metrics become less meaningful.

► Sparsity:

- One-hot vectors are sparse, leading to inefficiencies in storage and computation.
- Dense vectors mitigate this but still require large datasets for effective training.

► Lack of Context:

- Traditional VSMs do not capture word context effectively.
- Same word can have different meanings in different contexts (polysemy).

► Semantic Limitations:

- Cannot capture complex relationships like negation or antonymy.
- Similar words may not always be semantically related (e.g., "bank" vs. "river bank").

► Scalability:

- As vocabulary size increases, the term-document matrix becomes larger and more sparse.
- Requires significant computational resources for training and inference.

Summary

- ▶ **Contextual embeddings** (ELMo, BERT, GPT): Word vectors depend on sentence context.
- ▶ **Multilingual embeddings** and cross-lingual models.
- ▶ **Graph-based embeddings** (e.g., knowledge graph completion).
- ▶ **Hybrid embeddings**: Combining structured and unstructured data.

- ▶ Vector Space Models (VSMs) are foundational for modern NLP.
- ▶ Word embeddings capture semantic relationships in a continuous space.
- ▶ Word2Vec, GloVe, and fastText are key methods for generating word vectors.
- ▶ Dense embeddings outperform one-hot vectors in capturing meaning and relationships.
- ▶ Future work focuses on contextual, multilingual, and hybrid embeddings.

References

- [1] Mikolov, T., Chen, K., Corrado, G., & Dean, J. (2013).
Efficient Estimation of Word Representations in Vector Space.
arXiv:1301.3781.
- [2] Pennington, J., Socher, R., & Manning, C. D. (2014).
GloVe: Global Vectors for Word Representation.
EMNLP.
- [3] Bojanowski, P., Grave, E., Joulin, A., & Mikolov, T. (2017).
Enriching Word Vectors with Subword Information.
TACL.
- [4] Jurafsky, D., & Martin, J. H. (2023).
Speech and Language Processing (3rd Ed Draft).

- [5] Stanford NLP Slides.
<https://web.stanford.edu/class/cs224n>

- [6] Facebook AI Research (FAIR) – fastText.
<https://fasttext.cc>

- [7] Goldberg, Y. (2016).
A Primer on Neural Network Models for NLP.
JMLR.

- [8] Chrupała, G. (2019).
Symbolic and Subsymbolic Representations in NLP – Deep Learning Lectures.